TITLE PAGE

- **Problem Statement ID – <fill in>**
- **Problem Statement Title – SAAR: an inclusive, private AI digest for the community WhatsApp groups people actually live in**
- **Theme – Social Innovation & Inclusive Development**
- **PS Category – Software**
- **Team ID – <fill in>**
- **Team Name – <fill in>**

500 unread messages → 5 lines you can act on

School Parents Group

487 unread

forwards · stickers · good-morning images · 6 voice notes · one fee deadline buried at message 213

Scrolling is a privilege of time.

TODAY'S DIGEST

1. **Fee of ₹1,200 due Fri 5 pm. [12]**
2. **Sports day moved to 14 Aug. [31]**
3. **Bring Aadhaar copy Monday. [44]**

⚠️ **Fake "scholarship" link — ignore. [58]**

Also delivered as a 40-second voice note in the user's own language.

Runs silently as a linked device — on the user's own phone or a ₹0 cloud box.

No bot in the group · no admin permission · no new app to learn · data stays on the device

WORKING PROTOTYPE ALREADY RUNNING

Ingestion → local SQLite store → chunked map-reduce summariser → digest in self-chat → dashboard → jump-to-message extension

▶️ **Demo video link** 🔗 **Detailed flowchart**

WHY THIS IS SOCIAL INNOVATION

WhatsApp is India's real notice-board — for schemes, school, crops and health. Today its value goes to whoever has the time, literacy and eyesight to scroll. SAAR removes that filter, without asking anyone to change the app they already trust.

SAAR
PROPOSED SOLUTION

What is SAAR?

Where India's real notice-board meets AI

A silent AI assistant that reads the WhatsApp groups a person already belongs to, and hands back a short, cited, spoken digest of what actually matters to them.

Built for the 500 mn Indians whose civic life runs on WhatsApp.

How It Addresses the Problem

Panchayat notices, scheme deadlines, school fees, crop advisories and job leads all arrive in groups that push 300+ messages a day.

Catch-up digest
Summarises only what arrived since you last caught up.

Spoken, not just written
Voice-note digests in 22 languages — literacy stops being the gate.

Action, not just information
Dates, fees and documents become reminders you cannot miss.

Only what concerns you
A priority filter pings you when your name, child, crop or scheme comes up.

Trust by citation
Every line links to the real message, so nothing is taken on faith.

KEY FEATURES

1. **Catch-up digest** — Only what you missed
2. **Voice-note digest** — Text-to-speech, your dialect
3. **22 Indian languages** — Reads Hinglish and code-mix
4. **Voice notes read too** — Speech-to-text on group audio
5. **Action extractor** — Dates, fees, documents
6. **Scam & rumour shield** — Flags fraud links and forwards
7. **Priority alerts** — Ping only if it concerns you
8. **★ Save-to-vault** — Circulars kept after chat clears
9. **Ask my groups** — "When is the fee due?"
10. **Cited digests** — Tap a line, see the message
11. **Offline mode** — On-device model, zero cloud
12. **Caregiver relay** — Digest for elders, one tap

WHAT ALREADY WORKS TODAY

Live + historical message ingestion from every group, into a local encrypted SQLite store.

Chunked map-reduce summariser with [n] citations mapped back to real message ids.

Commands typed in your own chat: /sum /unread /topics /saved /schedule.

★ save-to-vault, daily scheduler, PWA dashboard and jump-to-message extension.

Innovation & Uniqueness

Zero-onboarding by design
Lives inside WhatsApp's own "Message Yourself" chat. No download, no sign-up, no new screen to learn — the reason it can reach people an app never would.

Invisible to the group
A linked device, not a bot. No admin rights, so no gatekeeper can decide who gets to be included.

Local-first and consent-first
Runs on the user's own device with an on-device model. Their chats never become someone else's dataset.

Cite-or-omit summarisation
Any claim the model cannot trace to a real message is dropped, not printed.

Who It Includes

Parents in school groups who cannot scroll 400 messages a day

Farmers and SHG women following advisory, mandi and loan groups

Low-literacy and first-time smartphone users — the digest is spoken

Blind and low-vision users — a screen-reader-first dashboard

Elderly members — a family caregiver can relay their digest

ASHA, anganwadi and panchayat workers juggling 20+ groups

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GROUP DIGEST

TECHNICAL APPROACH

1 · CAPTURE	2 · INGEST	3 · MULTILINGUAL INTAKE	4 · LOCAL STORE	5 · SUMMARISE	6 · AI LAYER	7 · DELIVER & ACT
User's own WhatsApp account, on their phone	Live messages (messages.upsert)	Text, type, sender and timestamp parsed	SQLite on device WAL · encrypted at rest	Window: since last catch-up, or a date range	Model router	Digest into the user's own Message-Yourself chat
One-time QR scan → Linked Device	History sync of older messages	Voice note → text (Whisper / Bhashini ASR)	messages · chats	Token-aware chunking	Offline: Ollama / Gemma on the device	Spoken digest (TTS) for low-literacy users
Baileys v7 socket (WhatsApp Web protocol)	Edits · deletes · reactions tracked	Image · PDF → OCR (circulars, forms)	catch-up markers · topic profiles	MAP: summarise each chunk	Online: Claude / GPT for long context	Priority alert if name, child, crop or scheme cited
Session keys stored locally in auth/	★ reaction by user → media downloaded	Language detect: 22 languages + Hinglish	★ vault: docs and media kept after chat clears	REDUCE: merge into one digest	Grounded prompt — cite or omit	Deadlines become reminders
No bot joins the group. No admin rights needed.	Sender + group metadata (JIDs)	Code-mix normalise and transliterate	vector index for "Ask my groups"	Extract actions, dates, fees, documents	Self-check pass drops uncited claims	Dashboard PWA — screen-reader first
Never marks as read — phone alerts stay intact.	Only groups the user opts in, one by one.	Translate on demand (IndicTrans2 / NLLB)	Nothing uploaded. No member is profiled.	Classify scam links and viral rumours	Reading level tuned to the user	Tap a citation → jump to the real message
Runs on a phone, an old laptop or a ₹0 cloud box	Read-only ingest — zero messages sent out	Local glossary: scheme, crop and exam terms	One tap erases a group's stored history	Cite each line as [n] → message id	Cost: about ₹1 per group per day	Daily or weekly digest, on a schedule

FEEDBACK LOOP · user rates a digest → catch-up marker advances, topic profile and priority filters retrain → the next digest is shorter, sharper and more personal

Runtime	WhatsApp	Data	AI	Bharat stack	Delivery
Node.js 22 · TypeScript strict ESM	Baileys v7 linked device · no Business-API fee	better-sqlite3 (WAL) · local-first · encrypted	Claude / GPT cloud · Ollama-Gemma offline · Whisper	Bhashini ASR + TTS · IndicTrans2 · 22 languages	PWA dashboard (node:http) · Chrome MV3 extension

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FEASIBILITY AND VIABILITY

TECHNICAL FEASIBILITY

1. Working prototype already runs: ingestion, local store, chunked map-reduce summariser, self-chat commands, dashboard.
2. Linked-device pairing is a one-time QR scan — no root, no custom ROM, no app install.
3. Summariser is model-agnostic; a 4 GB on-device model is enough for daily digests.
4. Speech and translation ride on proven stacks: Whisper, Bhashini, IndicTrans2.
5. Runs on a ₹0 cloud tier, an old laptop or a Raspberry Pi.

OPERATIONAL FEASIBILITY

1. The interface is a chat the user already opens daily — training cost is near zero.
2. One volunteer or ASHA worker can set up devices for a whole community in an afternoon.
3. Works on 2G-grade connectivity: a digest is a few KB of text and a short audio clip.
4. Degrades gracefully — if the model is unreachable, plain catch-up lists still arrive.
5. Scales one group at a time; no institution has to adopt it first.

ECONOMIC FEASIBILITY

1. No WhatsApp Business API cost, no per-message fee, no server per user.
2. Cloud model spend is about ₹1 per group per day; offline mode costs ₹0.
3. Free forever for individuals; schools, NGOs and panchayats fund the hosted tier.
4. One shared community box can serve a whole village or ward.
5. Open-source core keeps the cost of inclusion off the user.

LEGAL, ETHICAL & SOCIAL

1. Only the user's own account and own messages — no scraping of third parties.
2. Consent-first: summarisation is opt-in per group, and members are never profiled.
3. DPDP Act 2023 aligned by design: local storage, purpose limits, one-tap erase.
4. No content is ever sent back into a group, so no one can be impersonated.
5. Institutional deployments can migrate to the official WhatsApp Cloud API.

CHALLENGES → HOW WE OVERCOME THEM

Unofficial WhatsApp protocol may change or rate-limit	Never send to groups, never mark read, human-like pacing. An adapter layer lets us swap to the official Cloud API without touching the app.
A summary could distort or invent a decision	Cite-or-omit: uncited lines are dropped before delivery. Every digest links to the real message, and users rate each one.
Privacy of the other group members	On-device processing, no member profiling, no export of anyone's text. In offline mode chats never leave the phone.
Code-mixed and dialect speech is hard to transcribe	Bhashini and IndicTrans2 for Indian languages, plus a community-editable glossary of local scheme, crop and exam terms.
Low-end phones and patchy data	Digests are a few KB plus a short audio clip; the heavy work runs on a shared community box or a free cloud tier.

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GROUP DIGEST

IMPACT AND BENEFITS

What the user actually gets	SAAR	Scrolling it yourself	A group bot / assistant	Built-in chat AI
Effort to catch up on 400 messages	**One line to read or hear**	20–40 minutes	Group must add the bot	Manual, one chat at a time
Works for a low-literacy user	**Yes — spoken digest**	No	No — text only	No — text only
What the group members see	**Nothing at all**	—	Bot is visible to all	Varies
Needs admin permission	**No**	No	Yes	No
Data leaves the device	**Optional — offline mode**	No	Yes, to the bot host	Yes, to the vendor
Every claim traceable to a message	**Yes, cited [n]**	You are the source	Rarely	Rarely

Social

A parent, farmer or SHG member stops missing the deadlines that decide money, school and health.

Literacy and eyesight stop being the price of admission — the digest is spoken and screen-reader ready.

Scam links and viral rumours are flagged before they reach first-time digital users.

Economic

Missed fee, subsidy and scheme deadlines turn into money kept instead of money lost.

Frontline workers reclaim close to an hour a day from scrolling 20 groups.

Zero licence and zero per-message fee — one NGO box can cover a whole village.

Environmental & Systemic

Small on-device models replace repeated cloud calls, cutting compute and energy per digest.

Fewer "kya hua?" re-forwards means less network traffic and less storage churn.

The ★ vault keeps one copy of a circular instead of it being re-shared fifty times.

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RESEARCH AND REFERENCES

Evidence base and prior art — every title below is a live link

The problem, in data

› IAMAI–Kantar — Internet in India 2024

› NFHS-5 2019–21, IIPS / MoHFW — India report

› Garimella & Tyson — WhatsApp group data, ICWSM 2018

› Garimella & Eckles — WhatsApp misinformation, 2020

› Reis et al. — limiting message forwarding, 2019

Summarising real conversation

› Gliwa et al. — SAMSum dialogue corpus, 2019

› Zhong et al. — DialogLM, long dialogue, AAAI 2022

› Wu et al. — recursive map-reduce summarising, 2021

› Feng et al. — dialogue summarisation survey, 2022

› Lewis et al. — RAG, NeurIPS 2020 (Ask my groups)

Speech, language and access

› Radford et al. — Whisper speech-to-text, ICML 2023

› Gala et al. — IndicTrans2, all 22 languages, 2023

› Zhang et al. — PEGASUS summarisation, ICML 2020

› W3C — WCAG 2.2 accessibility guidelines

› MeitY — Digital Personal Data Protection Act 2023

ADOPTION & SUSTAINABILITY

- Free forever for individuals.
- Community box for schools, NGOs and panchayats: ₹499–₹4,999 a month.
- Open-source core — any state or NGO can self-host and audit it.

VALIDATION PLAN

- 8-week pilot: 3 school, 2 SHG and 1 panchayat group.
- Measured: deadlines caught, minutes saved, trust rating.
- Tested with screen readers and with non-reading users.

STANDARDS & COMPLIANCE

- DPDP Act 2023 — consent, purpose limits, erasure.
- WCAG 2.2 AA dashboard; RPwD Act 2016.
- No outbound message to any group, ever.

BUILT ON (open links)

- Baileys — WhatsApp Web library
- Ollama — on-device models
- Bhashini — Indian-language APIs
- WhatsApp Cloud API — migration path

Every line of a SAAR digest is traceable to a real message.

Inclusion without asking anyone to trust a black box — or to leave the app they already use.